10-725/36-725: Statistical Learning **Spring 2024**

Semi-supervised inference with machine learning model

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1 Introduction

In modern data analysis, we may encounter a situation that there are many unlabeled data, but the labeled data are very scarce, probably only a few hundred. Scientists want to use these data to draw scientific conclusion, such as protein activity, forest coverage rate etc. Nowadays, an increasing trend is to leverage machine learning model to fully utilize large amount unlabeled data, i.e. to impute labels. However, the predictive model can be inaccurate and naively treat the predictions as true label can lead to wrong conclusion. So there should be a debiased step when we use these prediction models. Recently, (Angelopoulos etc., 2023) proposes a method which can give correct inference based on any black-box machine learning model, which is called prediction-powered inference. And we will briefly review the idea below.

Notation $\mathcal{X} = \{(X_1, Y_1), (X_2, Y_2), \cdots, (X_n, Y_n)\}, \quad \mathcal{Y} = \{\tilde{X}_1, \tilde{X}_2, \cdots, \tilde{X}_N\}$ denotes labeled and unlabeled data, and they are i.i.d samples from a joint distribution $\mathbb{P}_{X,Y}$ and marginal distribution $\mathbb{P}_X$, respectively. Suppose we have a machine learning model f which predict label Y from X.

2 Method

Suppose we want to estimate the mean response $\mathbb{E}[Y]$, the most common estimator is sample mean $\hat{\theta}_{mean} = \frac{1}{n}\sum_{i=1}^{n} Y_i$, which is unbiased. However, in semi-supervised settings, there are huge amount of data which give us more information about the marginal distribution of X. If X and Y do have some relations, then we expect that leveraging $\tilde{X}$ can improve the estimation accuracy of Y.

The main idea of prediction powered inference is that we can add a debiased term to correct the bias caused by imperfect prediction:

$$\hat{\theta}_{PP} = \frac{1}{N}\sum_{i=1}^{N} f(\tilde{X}_i) + \frac{1}{n}\sum_{i=1}^{n}(Y_i - f(X_i)) \tag{1.1}$$

$$= \hat{\theta}_{mean} + \frac{1}{N}\sum_{i=1}^{N} f(\tilde{X}_i) - \frac{1}{n}\sum_{i=1}^{n} f(X_i) \tag{1.2}$$

Note that $\tilde{X}_i$ and (X_i, Y_i) are unlabeled and labeled data. The first term in (1.1) is just the sample mean of impute labels, and the second term is to estimate the prediction bias based on labeled data. It is easy to see that $\hat{\theta}_{PP}$ is unbiased because X and $\tilde{X}$ are from same marginal distribution $\mathbb{P}_X$:

$$\mathbb{E}[\hat{\theta}_{PP}] = \mathbb{E}[f(\tilde{X})] + \mathbb{E}[Y] - \mathbb{E}[f(X)] = \mathbb{E}[Y]$$

If the estimation is quite accurate, i.e. $f(X_i) \approx Y_i$, then we can expect the second term in (1.1) will close to zero. And because the number of unlabeled samples N is much larger than labeled sample n, there will be variance reduction compared to traditional estimator. Note that the unbiasness holds for any model f, whenever it is accurate or not.

Moreover, prediction powered estimator is just the sum of independent variables. By CLT, we have asymptotic normality: $(\theta \equiv \mathbb{E}[Y])$

$$\frac{\sqrt{n}}{\sqrt{\frac{n}{N}\sigma^2 + \sigma_\delta^2}}(\hat{\theta}_{PP} - \theta) \xrightarrow{d} \mathcal{N}(0,1)$$

where $\sigma^2 = \mathrm{Var}(f(X)), \sigma_\delta^2 = \mathrm{Var}(Y - f(X))$ and can be estimated by sample variance.

Then we can form confidence interval based on $\hat{\theta}_{PP}$ or $\hat{\theta}_{mean}$, respectively: $(\hat{\sigma}_Y^2 \equiv \frac{1}{n-1}\sum_{i=1}^n (Y_i - \hat{\theta}_{mean})^2)$

$$\hat{\theta}_{PP} \pm z_{1-\alpha}\sqrt{\frac{\hat{\sigma}^2}{N} + \frac{\hat{\sigma}_\delta^2}{n}}, \qquad \hat{\theta}_{mean} \pm z_{1-\alpha}\sqrt{\frac{\hat{\sigma}_Y^2}{n}}$$

We will compare the coverage rate and width of two CIs in the next section.

3 Experiment

We conduct a simulate experiment with synthetic dataset. We use a linear data generation model which adapts from (Zrnic & Candes, 2023): $X \sim \mathcal{N}(0, I_2)$, $Y = \mu + X^T\beta + \epsilon$, where $\mu = 4, \sigma^2 = 4$ with a varying $R^2 = \frac{\mathrm{Var}(X^T\beta)}{\mathrm{Var}(Y)} = \{0, 0.5, 1\}$. And $\beta_1 = \beta_2 = R\sigma/\sqrt{2}, \epsilon \sim \mathcal{N}(0, \sigma^2(1 - R^2))$ are set to keep the $\mathrm{Var}(Y) = \sigma^2 = 4$ constant when R^2 changes. The R^2 controls the degree to which outcomes can be explained by covariants. When $R^2 = 0$, $\beta_1 = \beta_2 = 0$, there is no relationship between X and Y. When $R^2 = 1$, $\epsilon = 0$, Y has become a deterministic function of X.

We individually generate 10000 unlabeled samples, 100 labeled samples and 100 extra labeled samples under each model. The extra samples are used to train the predictive model. It is noteworthy that when we use pre-trained model to form CI, it is unfair to compare two method because classical CI don't access to extra data. If we don't have a model in advance, a straightforward way is to split labeled data into training and inference part, but it will reduce the sample size. We briefly discuss this setting later. But first we consider situation where exists a model beforehead.

We use linear regression and random forest as our predictive models. For simplicity and consideration of computational cost, we choose the hyperparamters of random forest (number of trees, depth of tree, number of random features used in each tree) by single train-test split, and choose the best one which have the minimum MSE on test data. Figure 1.1 shows the mean and standard deviation of three different CI width under different signal-noise ratio. As we can see, the classical CI only use labeled data so it is not affected by the relations of X and Y. When Y is strongly correlated with X, the predictive model can significantly reduce CI width, which is more informative in real application. And in this case, linear regression do a better job than random forest because the true underlying model is linear.

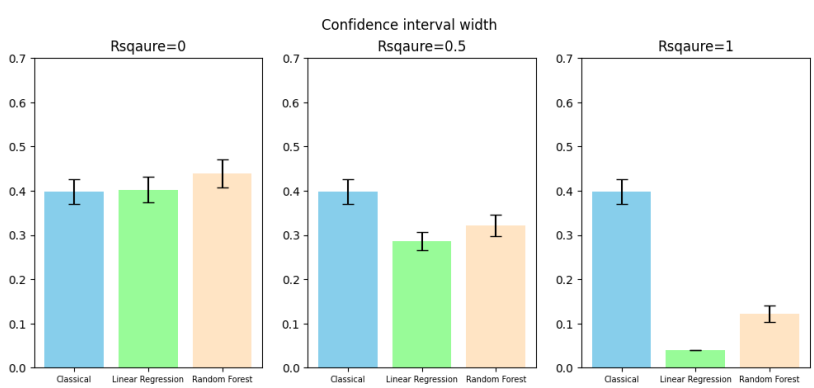

Figure 1.1: Confidence Interval Width

Figure 1.2 is the power function of three method varying with R^2. The true response mean is 4, so the first left point represent type I error rate of three estimator using z-test ($\alpha = 0.05$). Three estimator both near 0.05 thus type I error is controlled. And again prediction can improve test power if it is accurate. The more

accurate prediction is, the higher power can be reached.

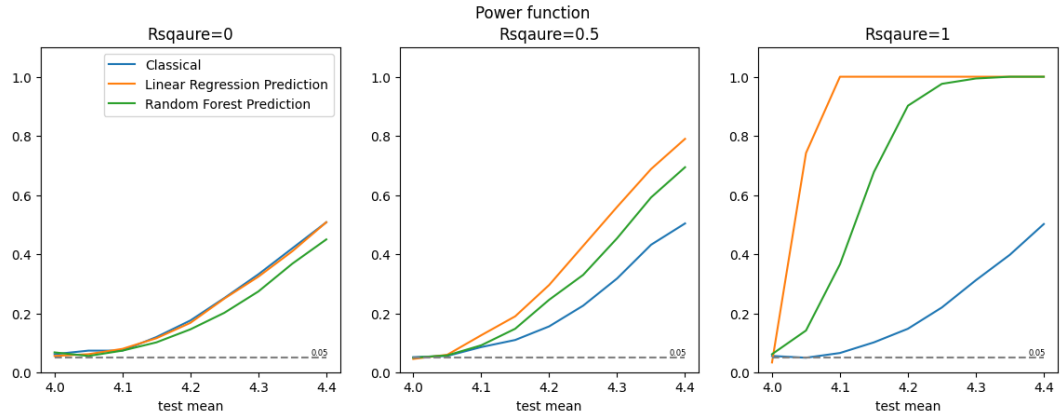

Figure 1.2: Power Function

As mentioned before, our predictive model has existed before conducting data analysis, and it is uncommon in practice. Therefore, we should compare the classical method with prediction powered method where the latter one need to split some data to train a model. And because of data splitting, it is not necessary that prediction powered estimator outperform than classical method, depending on whether trained model can predict unseen data well. Figure 1.3 shows the results with half-half split.

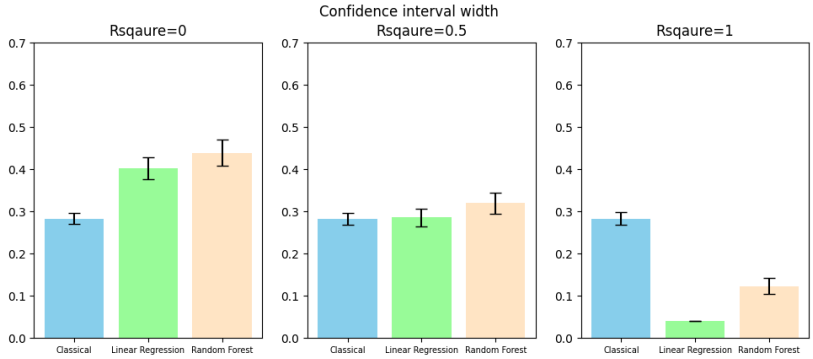

Figure 1.3: Confidence Interval Width Without Pre-trained Model

And it is reasonable to consider a more complex data generation model, such as a non-linear one. But what matter here is the prediction accuracy, using a non-linear model just increase fitting difficulty thus reduce accuracy. And we have manipulate it through R^2 in the linear model. As in the left plot of figure 1.1, an overfitting model can even impair estimation accuracy (random forest CI is slightly longer than other two CIs). When the model always give an opposite prediction ($f(X) = -Y$), it can have a much longer CI than classical CI.

Reference

[1] Anastasios N Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I Jordan, and Tijana Zrnic. Prediction-powered inference. arXiv preprint arXiv:2301.09633, 2023.

[2] Tijana Zrnic, Emmanuel J. Candes. Cross-Prediction-Powered Inference. arXiv preprint arXiv:2309.16598, 2023.